

Problem Set 6

POLS W4732

Fall 2026

Due November 20 before the TA section

Instructions. For each formal claim, include two or three algebraic inequalities or best-response comparisons that establish it. Unless directly following notation from slides or the text of the problem, define your notation.

Artificial-intelligence policy. Students may not use generative-AI systems (e.g., ChatGPT, Claude, Gemini, Copilot, etc.) to formulate, derive, solve, explain, or write answers to this problem set. Course materials, ordinary calculators, and collaboration within the registered problem-set group are permitted. **Every group member is responsible for understanding and being able to explain the entire submission.**

Exercise 1. Consider the following modified version of the career concern model of elections. Instead of an unbiased media coverage, the incumbent enjoys a systematically more positive coverage by the media. Recall that in the model the incumbent's performance is given by

$$\pi_1 = \theta_I + e_1 + \varepsilon_1,$$

where θ_I is drawn from a normal distribution with mean m and variance σ_θ^2 , e_1 is the effort level chosen by the incumbent, and ε_1 is a random shock. We assume that the politician's payoff from being elected is B , and his cost of exerting effort is $c(e) = e^2/2$. We capture media bias by assuming that the shock ε_1 is distributed normally with mean $b > 0$ and variance σ_ε^2 (for simplicity, suppose that ε_2 is still zero-mean). The voter knows that the media is biased. Like in the baseline model, effort in the second period is zero so the voter is only interested in choosing the candidate with the highest ability.

1. Write down the incumbent's expected payoff as a function of his effort choice e_1 .
2. Intuitively, do you expect effort to be higher or lower than in the baseline case of unbiased media? Explain.
3. Let's look at the voter's problem. For a given level of effort e_1 , the difference between observed performance and effort is given by

$$\pi_1 - e_1 = \theta_I + \varepsilon_1$$

$\theta_I + \varepsilon_2$ is distributed normally with mean m , but in the first period the voter observes $\theta_I + \varepsilon_1$, which is drawn from a normal with mean $m+b$. Write down the mean of the voter's posterior about the incumbent's ability.

4. What is the resulting cutoff $\bar{\pi}$ above which the incumbent is reelected? How does it compare with the one in baseline model ($\bar{\pi} = m + e_1$)? Explain.
5. Write down the probability of winning for the incumbent $\Pr(\pi_1 \geq \bar{\pi} | e_1)$.
6. Write down the first order condition characterizing the incumbent's first period's optimal effort choice
7. Substitute the value of the cutoff π into the incumbent's optimal effort and solve for the first period equilibrium level of effort e_1^* . How does it compare with the baseline model seen in class?

Exercise 2. Consider the basic career concern model developed in class. There are two periods $t \in \{1, 2\}$ and three players: voter (V), incumbent (I), and challenger (C). Voter cares about performance: $\pi_t = \theta_J + e_t + \epsilon_t$, where θ_J is the ability of politician in office, e_t is his level of effort, and ϵ_t is a random shock component (e.g., luck), with $\epsilon_t \sim \mathcal{N}(0, \sigma_\epsilon^2)$ (i.i.d. each period). In addition, the voter receives a payoff κ when the incumbent is reelected. The voter's per-period utility is thus:

$$U_v(\pi_t, J) = \pi_t + \mathbb{I}_{\{J=I\}}\kappa \quad (1)$$

Politicians are office-motivated. They get a payoff $B > 0$ from holding office and 0 otherwise. Effort is costly for politician and the cost is represented by $c(e)$, which is increasing, convex, and twice differentiable. Politician's ability is unknown at the beginning of the game. However, it is common knowledge that $\tilde{\theta}_J \sim \mathcal{N}(m, \sigma_\theta^2)$. The timing of the game is as follows:

- Nature draws θ_C and θ_I .
- **Period 1:**
 - 1. Incumbent chooses e_1
 - 2. Voter observes π_1 chooses whether to reelect incumbent
- **Period 2:**
 - 1. Elected official chooses e_2
 - 2. Payoffs are realized

Our equilibrium concept is PBE.

1. Show that the voter reelects the incumbent if and only if: $\pi_1 \geq \pi^* \equiv e_1^a + m - \frac{\kappa}{\lambda}$, where e_1^a is the anticipated level of effort and $\lambda = \frac{\sigma_\theta^2}{\sigma_\epsilon^2 + \sigma_\theta^2}$.
2. Show that when $|\kappa|$ is small enough,¹ the incumbent's first-period level of effort e_1^* is implicitly defined by:

$$c'(e_1^*) = \frac{B}{\sigma} \phi\left(-\frac{\kappa}{\lambda\sigma}\right)$$

3. For simplicity, suppose that $c(e) = \frac{e^2}{2}$ and $B = \sigma_\theta = 1$. Provide conditions under which more precise information (i.e., lower σ_ϵ) increases first period effort.

¹We need this condition to satisfy the second-order condition.